

Term Project Proposal

Console Minesweeper

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Overview

This project is a console-based minesweeper game implemented in F# using .NET 10. The player clears the hidden grid of mines by opening safe cells and placing flags on suspected mines. The application utilizes a keyboard-controlled cursor for navigation rather than standard text-input strings. This leads to providing an efficient, fast and dynamic user interface within the terminal environment.

Requirements

1. Board Display: The application will display a grid within the console terminal. The terminal will update the grid dynamically in place, to maintain a static visual layout.
2. Cell Status Representation: Cells have 5 states, and each states will be represented as different specific characters. Unopened cells as (■), opened cells with mine near as numbers(1-8), empty cell as (□). Found mines will be printed as (F) and stepped mine as (*).
3. Movements: Users will select their cells using the directional arrow keys. The current activated cell will be visually distinguished with other representation, so players could move their cursor efficiently.
4. Mine Checking: Players will check their mine using keyboards. Pressing the Spacebar executes open, and F will put a flag to a cell.
5. Prevent Initial Game Over: Mines will be distributed randomly, so player might open a mine as initial opening. This game prevents this as making a board right after the user's first opening.
6. Chording: When player presses Spacebar on an already opened number, cell will automatically open all other unflagged cells near that cell. If the player uses this function with wrong flags, it will be game over.
7. Game Modes: The game provides two modes. First is Basic board. The size of the board is fixed, and provides non-guessing solvable board, so the players could have clear solution. Second board is 'Harder!' board. Players can custom the size and number of mines, and there may be cells that players must guess.
8. Termination: If the player opens the cell that have mine, the game will be terminated with "Game Over!" message. The remaining mines will be opened at once. If player opens all numbers successfully, game will print "You Win!" and terminate the game.